Q1

Imagine a flat playground spinner. The distance between the center and the edge of the spinner is $r = 3\,\mathrm{m}$. It is a perfectly flat, solid circular platform that spins horizontally.
(a) The spinner rotates through an angle of $\theta = 4\,\mathrm{rad}$. Calculate the actual distance you traveled if you sit at the edge.
(b) If it takes $t = 2\,\mathrm{s}$ to complete this rotation, calculate the angular velocity (ω) of the spinner.
(c) Calculate your linear speed.

Q2

Imagine you place a small coin on the very edge of the same spinner from Q1.
(a) Calculate the centripetal acceleration of the coin while it is turning on the spinner.
(b) Which direction does this centripetal acceleration point? Depict circle as the top view of the spinner, and denote the acceleration arrow.
(c) Suddenly, the coin slips! Which direction does the coin fly in the horizontal direction? Explain using Newton's 1st Law.

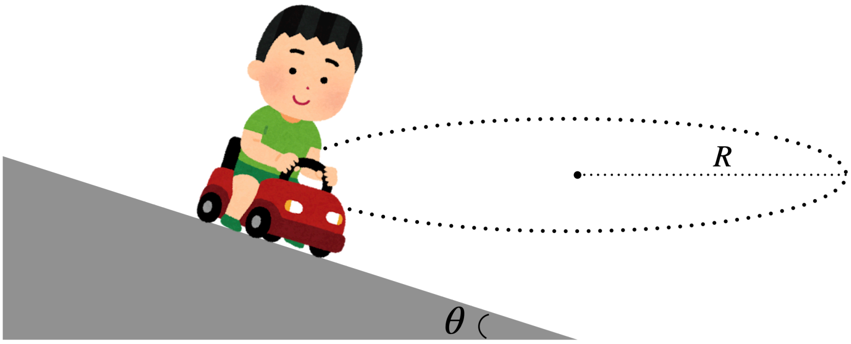
Q3

Imagine a $1200\,\mathrm{kg}$ car turning on a smooth, frictionless road that is banked (tilted) at an angle of $\theta = 37^{\circ}$. The car is making a HORIZONTAL circle.

(a) Calculate the exact value of the normal force.

(b) Calculate the centripital force (F_C).

(c) Calculate the centripetal acceleration (a_C) of the car.



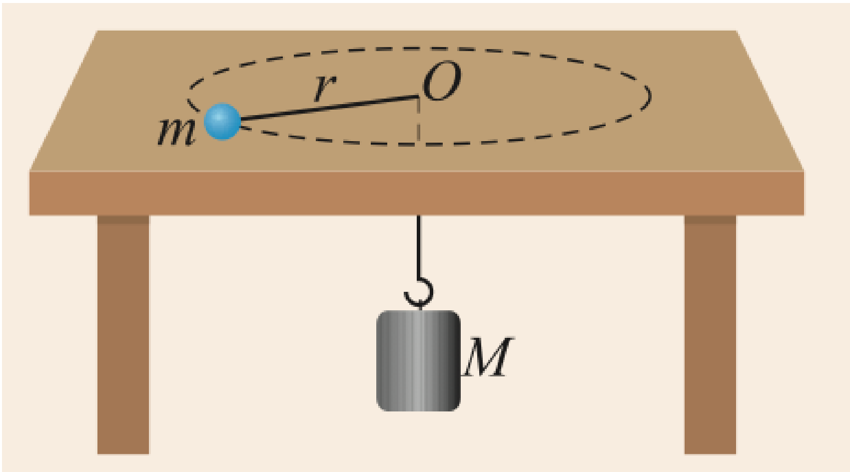
Q4

A small blue ball with mass $m = 5\,\mathrm{kg}$ is on a frictionless horizontal table. It is tied to a string that passes through a hole in the center to a hanging weight with mass $M = 4\,\mathrm{kg}$. The blue ball is swinging in a horizontal circle with a radius of $r = 2\,\mathrm{m}$.

(a) Calculate the Tension in the string.

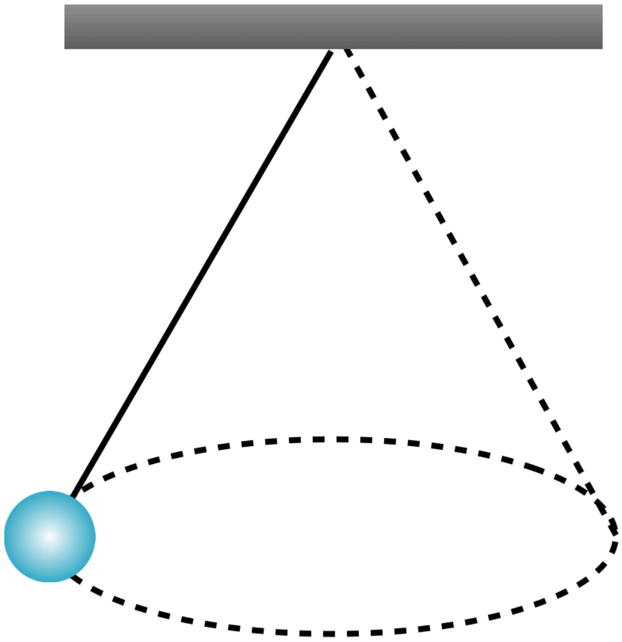
(b) Calculate the centripetal acceleration (a_C) of the blue ball.

(c) Calculate the blue ball's tangential speed.



Q5

A ball with a mass of $m = 1.2\,\mathrm{kg}$ is flying in a horizontal circle, suspended by a string of length $L = 1.5\,\mathrm{m}$. The string makes an angle of $\theta = 53^{\circ}$ with the vertical straight line.
(a) Calculate the exact value of the tension.
(b) Calculate the radius R of the horizontal circle and calculate the centripetal force.
(c) Calculate the tangential speed of the ball.



Q6

Analyze the Conical Pendulum further. Imagine you want the same toy plane to fly in a **WIDER circle** (meaning the angle θ becomes larger and the plane is higher up).
(a)Even if the circle is wider, the plane is still making a horizontal circle without falling. Does the vertical upward force (T_y) need to change? Explain your reason. (Hint: Look at Gravity).
(b) Because the angle θ is larger, the string is more "flat". To keep the SAME vertical force Ty while pulling at a flatter angle, must the total tension (T) in the string increase or decrease? Explain your reason.
(c) To maintain this new, wider circular path, does the plane need to fly faster or slower? Explain your reason.